Land 8: Sensorium Ruins

Senses and the autonomic system · OpenStax Ch. 14 to 16

Key ideas

Sensory receptors and general senses

- By stimulus: mechanoreceptors, thermoreceptors, chemoreceptors, photoreceptors, nociceptors (pain), osmoreceptors.
- By location: exteroceptors (outside), interoceptors (internal organs), proprioceptors (muscles, tendons, joints).
- Tactile (Meissner) corpuscles: light touch. Lamellated (Pacinian) corpuscles: deep pressure and vibration.
- Muscle spindles sense stretch; Golgi tendon organs sense tendon tension.
- Receptor signals become graded receptor potentials that trigger action potentials in sensory neurons.

Taste and smell

- Five tastes: sweet, salty, sour, bitter and umami (glutamate).
- Taste travels in VII (anterior two thirds of tongue), IX (posterior third) and X, to the solitary nucleus in the medulla, then thalamus, then gustatory cortex.
- Olfactory neurons are bipolar, replaced throughout life, and pass through the cribriform plate of the ethmoid bone.
- Smell synapses first in the olfactory bulb and reaches the cortex and limbic system without a first thalamic relay.
- Much of flavor depends on smell, so loss of smell makes food taste bland.

Hearing

- Outer ear: auricle, ear canal, tympanic membrane. Middle ear: malleus, incus, stapes; auditory tube to the nasopharynx.
- The stapes pushes on the oval window, making waves in cochlear fluid that move the basilar membrane; the round window releases the pressure.
- Hair cells of the spiral organ (organ of Corti) bend against the tectorial membrane, opening mechanically gated channels; signals go out cranial nerve VIII.
- Pitch: base (stiff, near oval window) for high tones, apex for low tones. Loudness: larger vibration and faster firing.
- Conductive loss: outer or middle ear problem (wax, otosclerosis). Sensorineural loss: hair cells or cochlear nerve.

Equilibrium

- Utricle and saccule (vestibule): maculae with otoliths sense head position and linear acceleration.
- Semicircular canals: three planes, cristae ampullares with a cupula sense rotation.
- Endolymph movement bends hair cell stereocilia; signals travel in the vestibular part of nerve VIII.
- Vestibular input is integrated with vision and proprioception in the brainstem and cerebellum.

Vision

- Fibrous tunic: sclera and cornea (most refraction). Vascular tunic: choroid, ciliary body, iris. Neural tunic: retina.
- Near vision: ciliary muscle contracts, ligaments slacken, lens rounds, pupil constricts. Presbyopia is age related loss of lens elasticity.
- Retina: photoreceptors to bipolar cells to ganglion cells; ganglion axons form the optic nerve at the blind spot (optic disc).
- Rods: dim light, gray scale, peripheral retina. Cones: color and sharp detail, packed in the fovea.
- Light turns retinal from cis to trans in rhodopsin; the rod hyperpolarizes and releases less glutamate.
- Nasal retina fibers cross at the optic chiasm; optic tract to thalamus (lateral geniculate) to occipital cortex. Chiasm lesion: loss of both temporal fields.

Somatosensory and motor pathways

- Dorsal column medial lemniscus: fine touch, vibration, proprioception; crosses in the medulla.
- Spinothalamic tract: pain and temperature; crosses in the spinal cord near the level of entry.
- Both pathways relay in the thalamus before reaching the postcentral gyrus.
- Corticospinal tract: upper motor neurons in the precentral gyrus; most fibers cross at the pyramids of the medulla.
- Upper motor neuron damage: spastic weakness, brisk reflexes, Babinski sign. Lower motor neuron damage: flaccid paralysis, lost reflexes, atrophy.

Reflexes

- Reflex arc: receptor, sensory neuron, integration center, motor neuron, effector.
- Stretch reflex (patellar): monosynaptic; muscle spindle excites the motor neuron of the same muscle, with reciprocal inhibition of the antagonist.
- Withdrawal reflex: polysynaptic response to pain, completed in the cord before pain is perceived.
- Crossed extensor reflex: the opposite limb extends to support the body during withdrawal.

Autonomic nervous system

- Two neuron chain: preganglionic neuron in the CNS, postganglionic neuron in a ganglion.
- Sympathetic: thoracolumbar (T1 to L2), short preganglionic fibers, chain and collateral ganglia; adrenal medulla releases epinephrine into blood.
- Parasympathetic: craniosacral (III, VII, IX, X and S2 to S4), terminal ganglia in or near organs; the vagus carries most of the output.
- All preganglionic neurons release ACh onto nicotinic receptors; parasympathetic postganglionic fibers release ACh onto muscarinic receptors; most sympathetic postganglionic fibers release norepinephrine.
- Alpha 1: vasoconstriction, pupil dilation. Beta 1: faster, stronger heartbeat. Beta 2: bronchodilation.
- Dual innervation of heart, iris, gut; sweat glands and most blood vessels are sympathetic only. Resting vagal tone slows the SA node from about 100 to about 70 per minute.

Drugs and the neurological exam

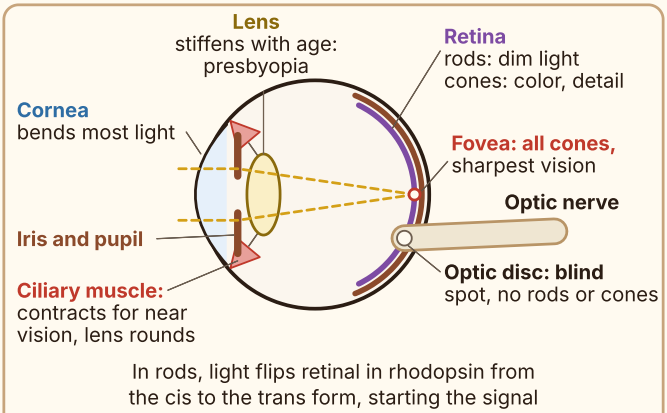
- Atropine blocks muscarinic receptors: fast heart, dry mouth, dilated pupils. Acetylcholinesterase inhibitors (organophosphates) cause the opposite.
- Beta blockers lower heart rate and contractility; albuterol (beta 2 agonist) opens airways; phenylephrine (alpha 1 agonist) raises blood pressure.
- Exam parts: mental status, cranial nerves, sensory, motor, reflexes, coordination and gait.
- Positive Romberg (falls with eyes closed) points to lost proprioception; cerebellar damage causes unsteadiness with eyes open and failed finger to nose testing.
- Babinski sign is normal in infants but signals corticospinal damage in adults.
- Eye down and out with ptosis and a dilated pupil: oculomotor nerve. Eye cannot look laterally: abducens nerve.

Villager wisdom

- Rods see by moonlight but only in shades of gray. Cones need bright sun and give us color and sharp detail.
- Sympathetic means fight or flight: racing heart, wide pupils, open airways. Parasympathetic means rest and digest: slower heart, small pupils, busy gut.
- Nicotinic receptors sit in every autonomic ganglion. Muscarinic receptors sit mainly on the organs that parasympathetic fibers reach.
- The semicircular canals feel your head spin, while the utricle and saccule feel tilting and moving in a straight line.
- Healers here chant LR6 SO4: the lateral rectus answers to the abducens nerve and the superior oblique to the trochlear nerve. The other eye movers follow the oculomotor nerve.

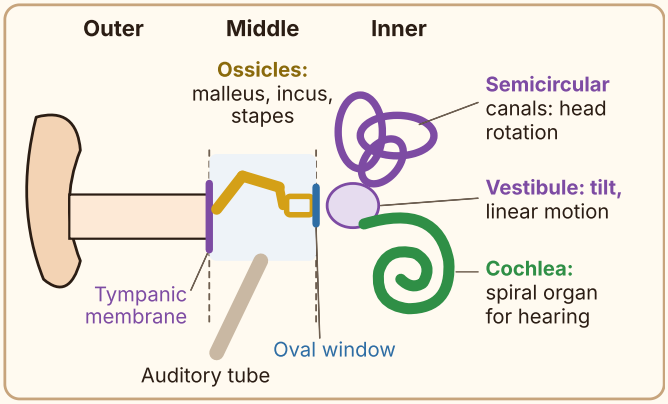
Diagrams

Anatomy of the eye



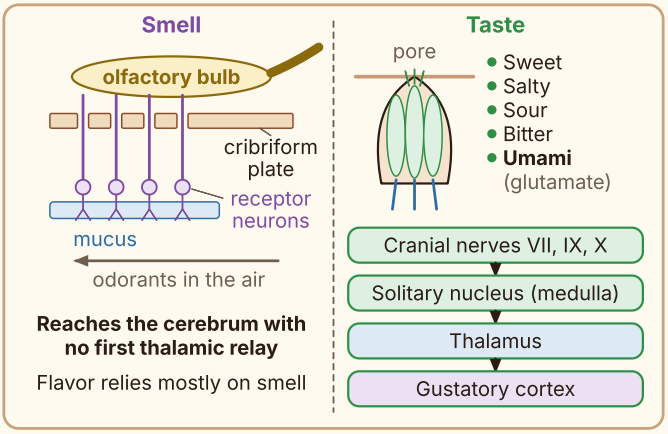
The cornea bends most incoming light and the lens fine tunes focus onto the fovea, where packed cones give the sharpest color vision; the optic disc has no photoreceptors, so it is the blind spot.

The ear: hearing and balance



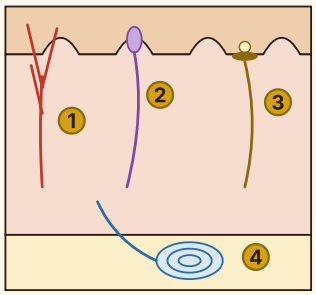
Sound vibrates the tympanic membrane and the ossicles (malleus, incus, stapes) pass it to the oval window and cochlea; the semicircular canals sense head rotation.

Smell and taste



Olfactory axons pass through the cribriform plate to the olfactory bulb and reach the cerebrum without a thalamic relay; taste travels in cranial nerves VII, IX and X to the medulla, thalamus and gustatory cortex.

Sensory receptor types



1 Free endings: pain, heat
2 Meissner: light touch
3 Merkel disc: fine touch
4 Pacinian: deep pressure, vibration

By stimulus

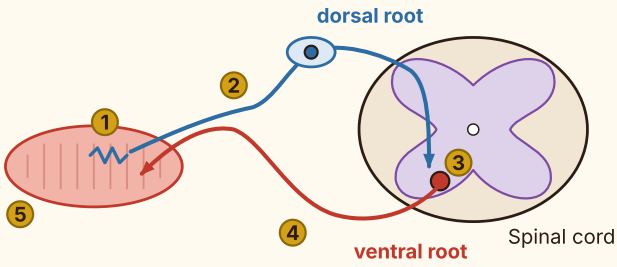
Mechanoreceptors
pressure, vibration, stretch
Thermoreceptors
temperature change
Nociceptors
tissue damage: pain
Chemoreceptors
chemicals: taste, smell
Photoreceptors
light: rods and cones

By location

Exteroceptors
outside stimuli, skin
Interoceptors
organs: bladder stretch
Proprioceptors
position: muscle spindles

Receptors are classed by stimulus (mechanoreceptors, thermoreceptors, nociceptors, chemoreceptors, photoreceptors) and by location (exteroceptors, interoceptors, proprioceptors such as muscle spindles).

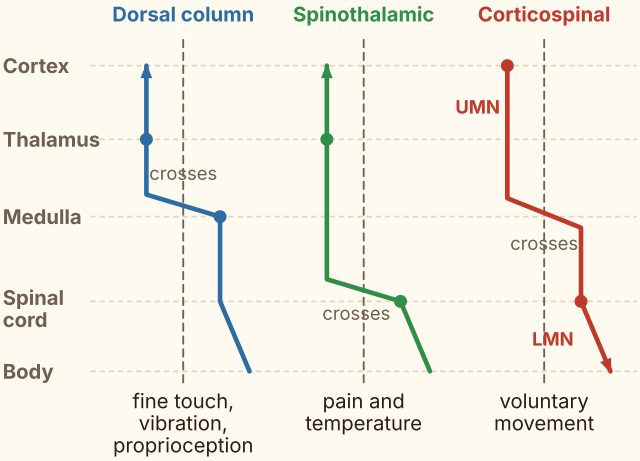
Reflex arc and spinal cord



1 Receptor (stretch)
2 Sensory neuron
3 One synapse in the gray matter
4 Motor neuron
5 Effector muscle

A reflex arc runs receptor, sensory neuron, integration center, motor neuron, effector; sensory fibers enter through the dorsal root and motor fibers leave through the ventral root.

Ascending and descending pathways



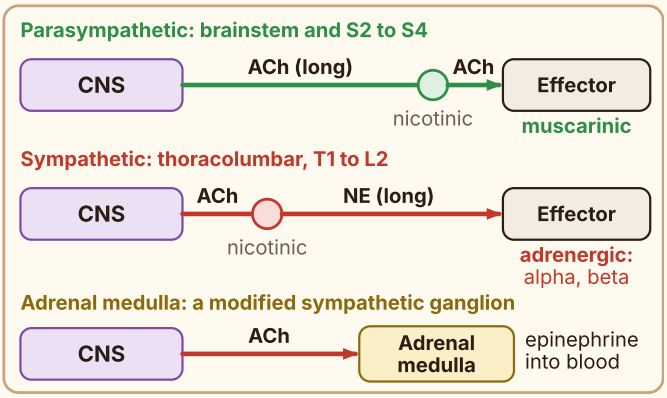
Dorsal column fibers cross in the medulla and spinothalamic fibers cross in the spinal cord, so a one sided cord cut loses pain on the opposite side; corticospinal upper motor neurons (UMN) cross at the pyramids and synapse on lower motor neurons (LMN).

Sympathetic versus parasympathetic

	Sympathetic	Parasympathetic
Role	fight or flight	rest and digest
Pupils	dilate	constrict
Heart rate	increases	decreases
Airways	dilate	constrict
Digestion	slows	speeds up
Origin	T1 to L2	brainstem, S2 to S4
Ganglia	near spinal cord	near or in organ
Effector NT	NE (mostly)	ACh

The sympathetic division prepares for fight or flight; the parasympathetic division conserves energy for rest and digest, and most organs receive both.

Autonomic pathways and receptors



All preganglionic neurons release ACh onto nicotinic receptors; parasympathetic postganglionic neurons release ACh onto muscarinic receptors, while most sympathetic ones release NE onto adrenergic receptors.

The twelve cranial nerves

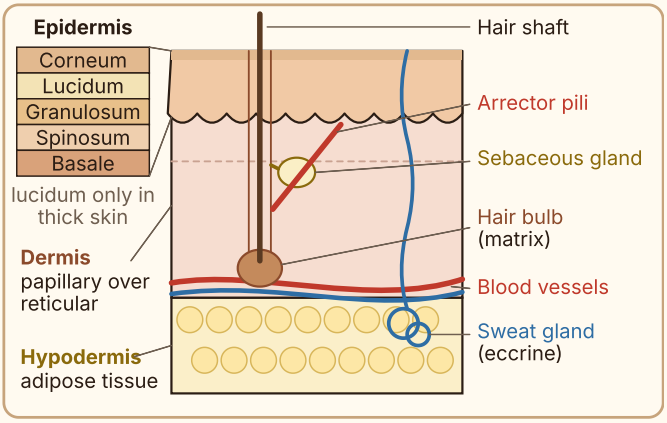
Sensory (blue), motor (red), both (purple)

- **I Olfactory** — smell
- **II Optic** — vision
- **III Oculomotor** — most eye moves, lid, pupil
- **IV Trochlear** — superior oblique muscle
- **V Trigeminal** — face sensation, chewing
- **VI Abducens** — lateral rectus: eye outward
- **VII Facial** — expression, eyelid, taste
- **VIII Vestibulocochlear** — hearing and balance
- **IX Glossopharyngeal** — taste, swallowing, gag
- **X Vagus** — heart, gut, swallow, voice
- **XI Accessory** — neck: SCM and trapezius
- **XII Hypoglossal** — tongue movement

XII lesion: the stuck out tongue deviates toward the injured side

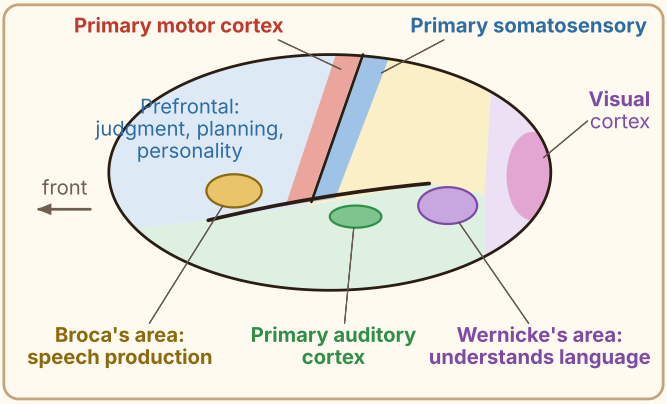
Cranial nerves I to XII are sensory, motor or mixed; for example the facial nerve (VII) closes the eyelid and makes facial expressions, and the abducens (VI) turns the eye outward.

Layers of the skin



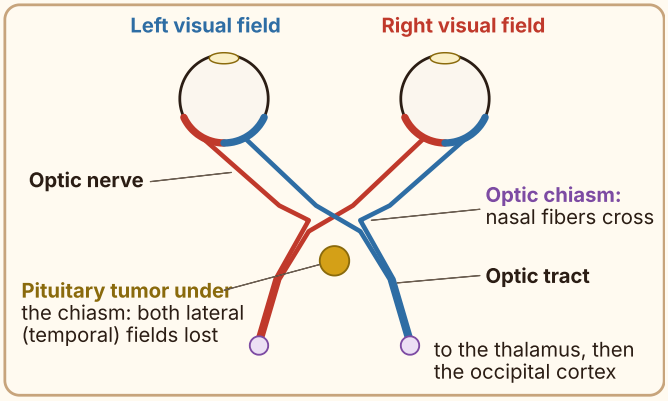
The epidermis is stratified squamous epithelium renewed from the stratum basale; the dermis holds glands, follicles and vessels; the fatty hypodermis insulates and cushions.

Functional areas of the cerebral cortex



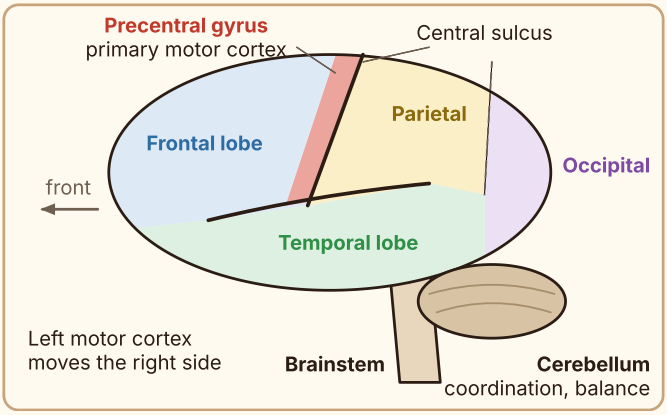
Broca's area in the frontal lobe plans the movements of speech, while Wernicke's area near the temporal lobe lets us understand language; the precentral gyrus moves the body and the postcentral gyrus feels it.

Visual pathway and the optic chiasm



At the optic chiasm only fibers from the nasal (medial) half of each retina cross, so a pituitary tumor pressing on the chiasm cuts both lateral (temporal) visual fields.

Regions of the brain



The precentral gyrus (primary motor cortex) controls the opposite side of the body, and the cerebellum coordinates smooth, accurate movement and balance.

Self check

1. **The adrenal medulla acts like a modified sympathetic ganglion because its chromaffin cells:**
    A. Receive postganglionic sympathetic fibers
    B. Release acetylcholine into the blood
    C. Are controlled by the vagus nerve
    D. Release epinephrine into the blood
2. **After the nucleus of the solitary tract in the medulla, taste signals relay through which structure on their way to the gustatory cortex?**
    A. Olfactory bulb
    B. Cerebellum
    C. Superior colliculus
    D. Thalamus
3. **Receptors that monitor internal organs, such as stretch of the bladder wall, are classified as:**
    A. Interoceptors
    B. Exteroceptors
    C. Proprioceptors
    D. Photoreceptors
4. **A heart transplant patient's donor heart has no autonomic nerves, and its resting rate is about 100 beats per minute. This shows that normally:**
    A. Vagal tone slows the SA node at rest
    B. Sympathetic tone slows the heart at rest
    C. The SA node cannot fire on its own
    D. Hormones slow the heart at rest
5. **The dorsal column medial lemniscus pathway carries:**
    A. Fine touch, vibration and proprioception
    B. Pain and temperature
    C. Voluntary motor commands
    D. Visceral motor output
6. **A 48 year old can see distant objects well but now needs reading glasses because the lens has lost its elasticity. This is:**
    A. Myopia
    B. Hyperopia
    C. Cataract
    D. Presbyopia
7. **A farm worker is poisoned by an organophosphate that blocks acetylcholinesterase. Which signs are expected?**
    A. Heavy salivation, slow heart rate, small pupils
    B. Dry mouth, fast heart rate, wide pupils
    C. Bronchodilation and warm, dry skin
    D. Decreased gut motility and urine retention

8. **The optic disc produces a blind spot because it:**
    A. Has no photoreceptors where the optic nerve exits
    B. Contains only cones
    C. Is covered by the lens
    D. Is packed with rods
9. **Receptors that respond to pressure, vibration and stretch are classified as:**
    A. Chemoreceptors
    B. Thermoreceptors
    C. Mechanoreceptors
    D. Photoreceptors
10. **Receptors that bind norepinephrine and epinephrine are called:**
    A. Muscarinic receptors
    B. Nicotinic receptors
    C. GABA receptors
    D. Adrenergic receptors
11. **A positive Babinski sign (big toe extends upward when the sole is stroked) is a normal finding in:**
    A. Healthy adults
    B. Infants under about one year
    C. Athletes after exercise
    D. Healthy older adults
12. **The semicircular canals of the inner ear mainly detect:**
    A. Rotation of the head
    B. Head tilt while standing still
    C. Linear acceleration in an elevator
    D. The pitch of sounds

Answer key

1. **D.** Release epinephrine into the blood. Preganglionic sympathetic fibers synapse directly on chromaffin cells, which act like postganglionic neurons and release mostly epinephrine into the bloodstream. No postganglionic fibers are involved.
2. **D.** Thalamus. Taste follows the usual sensory route through the thalamus before reaching the gustatory cortex. Only smell skips a first thalamic relay, so the olfactory bulb is not part of the taste pathway.
3. **A.** Interoceptors. Interoceptors (viscероceptors) sense conditions inside the body, such as organ stretch or blood chemistry. Proprioceptors also sense internal information, but specifically the position and movement of muscles and joints.
4. **A.** Vagal tone slows the SA node at rest. The SA node fires about 100 times per minute on its own, and resting parasympathetic (vagal) tone brings it down to about 70. Sympathetic input speeds the heart rather than slowing it.
5. **A.** Fine touch, vibration and proprioception. The dorsal columns carry discriminative touch, vibration and body position to the thalamus and somatosensory cortex. Pain and temperature travel in the spinothalamic tract.
6. **D.** Presbyopia. Presbyopia is the age related loss of lens elasticity that limits accommodation for near vision. Hyperopia also blurs near vision but is caused by a short eyeball or weak lens, not aging of the lens.

7. **A.** Heavy salivation, slow heart rate, small pupils. Without the enzyme, acetylcholine builds up and overstimulates muscarinic receptors, giving parasympathetic signs such as salivation, bradycardia and pinpoint pupils. Dry mouth and wide pupils are the opposite pattern, seen with atropine.
8. **A.** Has no photoreceptors where the optic nerve exits. At the optic disc ganglion cell axons leave the eye as the optic nerve, leaving no room for rods or cones. A spot with only cones is the fovea, the area of sharpest vision.
9. **C.** Mechanoreceptors. Mechanoreceptors open ion channels when they are physically deformed by touch, pressure, vibration or stretch. Chemoreceptors respond to dissolved chemicals instead, as in taste and smell.
10. **D.** Adrenergic receptors. Alpha and beta adrenergic receptors bind norepinephrine and epinephrine. Muscarinic and nicotinic receptors are cholinergic, binding acetylcholine.
11. **B.** Infants under about one year. In infants the corticospinal tract is not yet fully myelinated, so the Babinski response is normal. In anyone older it suggests upper motor neuron damage.
12. **A.** Rotation of the head. Fluid in the three semicircular canals lags behind when the head turns, bending hair cells in the ampulla and signaling rotation. Head tilt and straight line acceleration are sensed by the utricle and saccule instead.

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